

Supplementary Material

Shuo Yang, *Student Member, IEEE*, Zhengshuo Li, *Senior Member, IEEE*, Zhirou Wang, *Student Member, IEEE*

I. DEPENDENCE OF POWER FLEXIBILITY REGION BOUNDARY IDENTIFICATION ON THE INITIAL POINT

MOST existing studies that identify the power flexibility region (PFR) boundary via discrete angular scanning start from the origin or an interior point and scan over all directions to determine boundary points by maximizing the radial distance. However, the accuracy of such algorithms can strongly depend on the selected reference point. This dependence is illustrated by the example in Fig. R1.

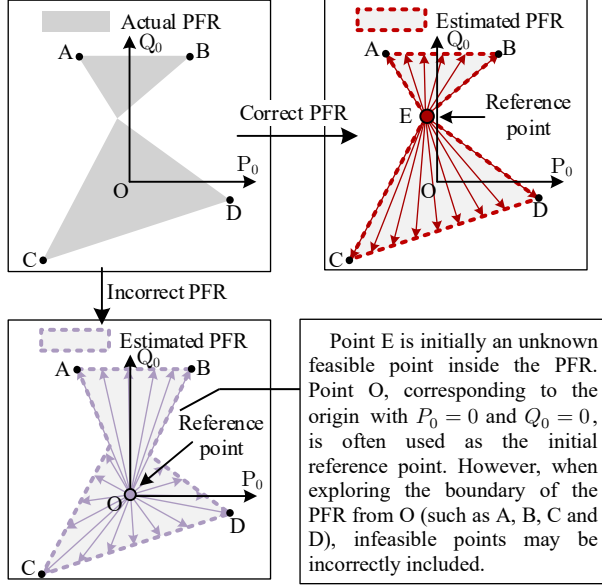


Fig. R1. Examples of PFR estimation.

In the example shown in Fig. R1, the true PFR is illustrated in Fig. R1(a). By choosing different reference points, typically the origin or a power-flow feasible point, and performing a full-angle scan, the two PFRs shown in Fig. R1(b) and (c) are obtained. It is evident that the identified PFR depends on the choice of the reference point and may be inaccurate.

II. THE PSEUDOCODE OF THE PROPOSED PFR IDENTIFICATION ALGORITHM

The complete procedure of the proposed PFR identification algorithm is given in Algorithm 1.

III. SUPPLEMENTARY DESCRIPTIONS OF THE TEST MODEL

This section provides supplementary descriptions of the relevant parameters of the modified IEEE 13-node, Swiss 38-node, and unbalanced 162-node test systems, as listed below.

A. Supplementary Description about the modified IEEE 13-node test system

Values of relevant regulating devices in the modified IEEE 13-node test system can be found in Table RI. Here, l_{min} is the gap threshold for terminating the line-search iteration; M is the number of scanning angles; ε is a numerical tolerance used in the algorithm; S_{Line} denotes the per-unit maximum

Algorithm 1: The proposed PFR Identification Algorithm.

Input: The number of scanning angles M ; an empty feasible/infeasible set Θ_{fe}/Θ_{inf} and a temporary feasible/infeasible set $\Theta_{fe}^m/\Theta_{inf}^m$; the gap for terminating the iteration l_{min} .
Output: The feasible set Θ_{fe} and the infeasible set Θ_{inf} .

```

1 for  $m = 0 : M$  do // This step can be executed in parallel.
2   Feasibility Initialization;
3   if the problem is infeasible, then
4     go to 22;
5   else
6     if  $l_B - l_A \leq l_{min}$ , then
7        $\Theta_{fe}^m \leftarrow \Theta_{fe}^m \cup \{X_A, X_B\}$  and break;
8     else
9       Feasibility Expansion;
10      if  $l_C \geq l_D$ , then
11        go to 22;
12      else
13         $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \cup \{X_C, X_D\}$ ; Feasibility Exploration;
14        if  $l_E = l_C + l_{tol}$ , then
15           $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \setminus \{X_C\}$ ; go to 25;
16        else
17           $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \cup \{X_E, X_F\}$ ; go to 25;
18        end
19      end
20    end
21  end
22   $\Theta_{fe} \leftarrow \Theta_{fe} \cup \Theta_{fe}^m$ ;  $\Theta_{inf} \leftarrow \Theta_{inf} \cup \Theta_{inf}^m$ ;
23   $m \leftarrow m + 1$ ;
24 end
25 if  $l_D - l_F > l_{min}$ , then
26    $(X_C, Y_C, Z_C) \leftarrow (X_F, Y_F, Z_F)$ ;  $l_D \leftarrow l_F$ ; go to 10;
27 elseif  $l_D < l_F$ , then
28    $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \setminus \{X_C\}$ ;
29 else
30   go to 22;
31 end

```

TABLE RI

VALUES OF RELEVANT PARAMETERS FOR 13-NODE SYSTEM (UNIT: P.U.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	1
M	120	v_{MV}^{max}	1.05	P_{load}	0
ε	0.001	P_{PV}^{max}	1	Q_{load}	0
S_{Line}	1	P_{PV}^{min}	0.2	$q_{C1/2/3/4}$	0/0.5/1/1.5
v_{OLTC}	0.98/1/1.02	$\tan \phi_{PV}$	1	S_c	1

TABLE RII

VALUES OF RELEVANT PARAMETERS FOR 38-NODE SYSTEM (UNIT: P.U.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	0.8
M	120	v_{MV}^{max}	1.05	$q_{RC1/2/3/4/5}$	-1/-0.5/0/0.5/1
ε	0.001	P_{PV}^{max}	0.8	S_c	1
S_{Line}	1.5	P_{PV}^{min}	0		
v_{OLTC}	0.98/1/1.02	$\tan \phi_{PV}$	1		

TABLE RIII

VALUES OF RELEVANT PARAMETERS FOR 162-NODE SYSTEM (UNIT: P.U.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	1
M	120	v_{MV}^{max}	1.05	S_{Line}	1
ε	0.001	P_{PV}^{max}	1	S_c	1
$q_{RC1/2/3/4/5}$	-1/-0.5/0/0.5/1	P_{PV}^{min}	0		
v_{OLTC}	0.98/0.99/1/1.01/1.02	$\tan \phi_{PV}$	1		

line capacity; v_{OLTC} denotes the set of voltage tap positions of the on-load tap changer; $v_{MV}^{min}/v_{MV}^{max}$ are the lower and upper nodal voltage limits, respectively; P_{PV}^{max} and P_{PV}^{min} are the upper and lower active-power limits of photovoltaic (PV) units, respectively; $\tan \phi_{PV}$ denotes the maximum tangent of the PV power-factor angle; S_{PV}^{max} denotes the maximum apparent-power capacity of distributed PV; P_{load} and Q_{load} are the active and reactive loads at each node; $q_{C1/2/3/4}$ represents the per-unit reactive-power compensation corresponding to different switching levels of capacitor banks; and S_c denotes the per-unit transformer capacity.

The line parameters of the modified IEEE 13-node test system used for testing can be found in [1].

B. Supplementary Description about the Swiss 38-node test systems and unbalanced 162-node test systems

The diagrams of the Swiss 38-node test system and unbalanced 162-node test system are shown in Fig. R2 and the values of relevant regulating devices in the two test systems can be found in Table RII and RIII.

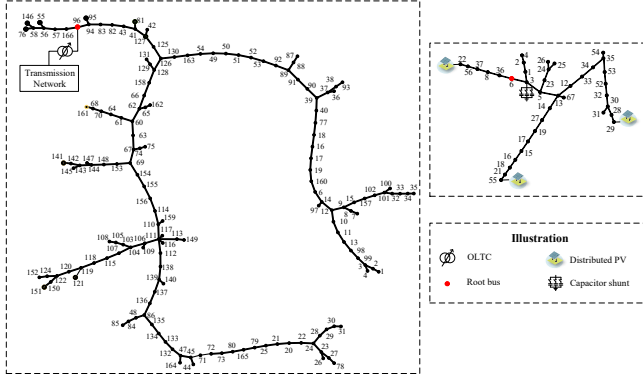


Fig. R2. Diagrams of (a) the modified 162-node test system and (b) the Swiss 38-node test system.

In Table RIII, $q_{RC1/2/3/4/5}$ denotes the reactive-power output levels of the reactor/capacitor bank. In addition, to better illustrate the nonconvexity of the PFR, the load level of the 38-node system is scaled by a factor of five.

The line parameters and load data of the 38-node Swiss test system and 162-node test system can be found in [2].

IV. THE PFR OF THE UNBALANCED MODIFIED SWISS 162-NODE TEST SYSTEM

The proposed method is further evaluated on the unbalanced modified Swiss 162-node test system. For comparison, Fig. R3 reports the estimated PFR obtained under the identical-three-phase assumption.

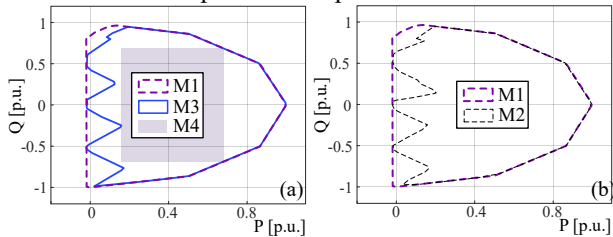


Fig. R3. PFRs obtained by different methods: (a) M1, M3, and M4 for the 162-node system; (b) M1 and M2 for the 162-node system.

V. SUPPLEMENTARY DESCRIPTION OF THE PERFORMANCE METRICS IN TABLE I

The intersection-over-union (IoU) is defined as the ratio of the overlap area between two regions to their total covered area. The infeasible inclusion ratio (IIR) and the feasible omission ratio (FOR) represent the proportions of infeasible operating points incorrectly included and feasible operating points incorrectly excluded, respectively.

Let A denote the estimated region obtained by a candidate method and B denote the benchmark region obtained by the enumeration method, and let $|\cdot|$ denote the area of a region. Then, the IoU, IIR and FOR are defined as (1)-(3):

$$IoU = \frac{|A \cap B|}{|A \cup B|}, \quad (1)$$

$$IIR = \frac{|A \setminus B|}{|B|}, \quad (2)$$

$$FOR = \frac{|B \setminus A|}{|B|}. \quad (3)$$

A larger IoU indicates higher geometric similarity. Smaller IIR and FOR values indicate fewer incorrectly included infeasible operating points and fewer incorrectly omitted feasible operating points, respectively.

VI. SUPPLEMENTARY DESCRIPTION OF THE PROPOSED PFR IDENTIFICATION ALGORITHM

A. Proof of Convergence of the PFR Identification Algorithm

For each scanning direction, the line-search procedure explores binary configurations along the corresponding ray. In Step 1, two boundary configurations are obtained by solving the minimum- and maximum-distance problems. If these two configurations do not provide a continuous feasible segment, Step 3 searches for the next feasible segment by imposing $l_m \geq l_{m,C} + \varepsilon$. This constraint prevents the algorithm from repeatedly returning the same feasible point within the prescribed tolerance. Since the number of binary configurations is finite, the number of newly explored configurations along a ray is also finite. Therefore, the line-search procedure terminates after a finite number of iterations. Let N_{01} denote the total number of binary configurations. The worst-case number of line-search iterations for each scanning direction is bounded by N_{01} , and the overall worst-case number of explored configurations is bounded by $O(MN_{01})$. In summary, the algorithm is guaranteed to converge within a finite number of iterations.

B. Sensitivity of the proposed algorithm to M , l_{min} and l_{tol}

In this section, the sensitivity of the proposed PFR identification algorithm to the number of angular partitions M , the gap for terminating the line-search iteration l_{min} , and the small positive tolerance l_{tol} is evaluated on the 13-node test system. In the following tables, the IoU always denotes the intersection-over-union between the PFR obtained by the proposed method and that obtained by the enumeration method. The results are summarized as follows.

We first assess the sensitivity of the proposed algorithm to M . The test scenario is identical to the 13-node test system

presented in the manuscript, except that only the value of M is changed.

TABLE RIV

SENSITIVITY OF THE PFR IDENTIFICATION ALGORITHM TO M						
M	15	30	60	120	240	360
IoU	0.875	0.955	0.993	0.995	0.995	0.995

It can be observed that as M increases, the IoU first increases and then tends to remain unchanged. This indicates that the estimation accuracy of the PFR generally improves with the increase of M . However, beyond a certain threshold, for example when $M = 120$, the improvement in PFR accuracy becomes marginal and the IoU remains almost unchanged. Therefore, in practical applications, M should be properly selected to achieve a trade-off between evaluation accuracy and computational cost.

Then, the sensitivity of the proposed algorithm to l_{min} and l_{tol} is evaluated. Except for the parameter under evaluation, all other settings are the same as those in the manuscript.

TABLE RV

SENSITIVITY OF THE PFR IDENTIFICATION ALGORITHM TO l_{min} AND l_{tol}		
l_{min}	l_{tol}	IoU
0.005	0.0001	0.995
	0.001	0.995
	0.01	0.995
	0.1	0.995
0.001	0.001	0.995
0.005		0.995
0.05		0.979
0.5		0.956

It can be observed that, in the test case, increasing l_{tol} has almost no impact on the IoU. This is because l_{tol} mainly affects the identification of interior infeasible points, whereas the tested case contains no interior infeasible points and only exhibits nonconvexity near the boundary. Therefore, the resulting impact is limited. In practical tests, it is still recommended that l_{tol} be set smaller than l_{min} . In addition, as l_{min} further increases, the IoU generally decreases. This is because l_{min} affects the identification of nonconvex boundaries and interior infeasible regions of the PFR. An excessively large l_{min} may cause feasible points to be incorrectly classified as infeasible during the PFR identification process, thereby reducing the IoU. Therefore, in the test case, a relatively small value of l_{min} , such as 0.005, is recommended.

REFERENCES

- [1] "IEEE13NodeSystem.dss." Accessed: Jun. 28, 2024. [Online]. Available: <https://github.com/YYY-maker130/Assessment-of-the-Security-Region.git>
- [2] A. E. Oneto, F. Tettamanti, B. Gjorgiev and G. Sansavini, "Synthetic low- and medium-voltage grids for Switzerland," *Zenodo*, Apr. 2025. doi: 10.5281/zenodo.15167589.